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Overcoming Frictional Limitation in Open Hole Acid Stimulation: Successful Deployment of Hydraulic Activated Friction Reduction Tool (HAFRT) in Highly Deviated Extended Reach Water Injector Wells in UAE

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Abstract

Highly deviated extended reach open-hole sections in water injector wells present significant challenges for coiled tubing (CT) operations, particularly during acid stimulation treatments, where tool reach and effective fluid placement are critical. Excessive drag and friction often limit CT reach, resulting in incomplete treatment coverage and suboptimal stimulation results. This paper presents a successful case study of Hydraulic Activated Friction Reduction Tool (HAFRT) to overcome these limitations in a challenging open hole acid stimulation application in two water injector wells in the UAE: Well X and Well Y.

The subject wells had a history of multiple unsuccessful CT intervention attempts due to high frictional forces and insufficient CT reach resulting in less effective stimulation treatments. In the latest operation, multiple runs were conducted – a) with conventional bottom hole assembly (BHA), b) with conventional BHA and acid spotting, and c) with HAFRT. HAFRT was deployed to introduce axial oscillations along the CT string, reducing contact between CT and wellbore while improving tool conveyance in the challenging open hole section. HAFRT transforms hydraulic energy into gentle oscillations that convert static friction into dynamic friction, extending CT reach beyond lock up point. Improved accessibility using HAFRT was followed by pin-point acid stimulation to improve injection which was not possible earlier due to limited accessibility.

These interventions resulted in significant reduction in friction, extending CT reach beyond previous lock up limits while enabling effective acid placement across the entire targeted open hole interval. HAFRT was activated from the beginning of the open hole section to delay lock up and overcome challenging hole condition without inducing buckling throughout the run, resulting in substantial CT weight improvement. Eventually, the acid stimulation operation was successfully completed with 100% coverage of open hole section in Well X as compared to previous operations. In Well Y, even though accessibility was improved compared to previous CT interventions, due to CT size limitations, 100% of the open hole could not be accessed. This is one of the lessons learned from this case study.

Accessibility improvement in Well X using HAFRT compared to without HAFRT was 92% with improvement in injectivity index (II) of 80%. On the other hand, in Well Y, accessibility improvement with and without HAFRT was 117% with II improvement of 5 folds. This paper explores the use of HAFRT in extended reach wells to improve accessibility which could not be accessed with conventional CT BHAs, thereby providing access to the extended section of the open hole for any treatment. Lessons learned from these case studies are also discussed in this paper, which can form the basis of future well intervention in similar extended reach wells.

Introduction

Acid stimulation is critical for long horizontal wells primarily to remove drilling fluids, debris, and any other damage that can restrict hydrocarbons, minimize well productivity or restrict wells fluid injection. In these wells, particularly extended horizontal sections, the accumulation of drilling fluids and solids in the wellbore face can significantly impede reservoir access and inhibit efficient production or injection.

Coiled tubing is preferred for acid stimulation applications due to its flexibility, continuous deployment capabilities, and efficient fluid placement. In long horizontal wells, where access can be challenging, coiled tubing offers the ability to navigate complex well geometries and reach extensive lateral sections without the need for multiple joints or connections. Its continuous length and seamless design allow for safer and quicker deployment into wells compared to jointed drill pipe string, reducing operational downtime and associated costs. Coiled tubing also enables precise control over the placement and concentration of acid treatments, optimizing reservoir contact and enhancing well productivity. Moreover, its ability to circulate fluids during operations helps in maintaining well integrity by minimizing formation damage and improving overall treatment effectiveness. These factors collectively make coiled tubing an indispensable tool for acid stimulation and wellbore cleanout applications ensuring enhanced reservoir performance while maximizing operational efficiency.

However, deploying coiled tubing in long horizontal wells has multiple challenges due to the nature of coiled tubing string and available operational resources. To overcome these challenges, Hydraulic Activated Friction Reduction Tool (HAFRT) is deployed along with the coiled tubing string to reduce frictional forces acting on the coiled tubing, eventually reducing the risk of fatigue and buckling. This is particularly crucial in highly deviated or extended horizontal wells where the tubing is subjected to significant tensile and compressive forces.

This paper will study the application challenges, resource limitations and use of flow activated friction reduction tools to overcome these challenges in long horizontal, extended-reach wells to perform acid stimulation and reservoir cleanout operations.

Application Overview

To perform acid stimulation in long horizontal and extended-reach wells deploying coiled tubing has become more challenging with the increasing complexity of well profiles. The limitation to deploying coiled tubing into relatively deep intervals has been challenging for conducting acid stimulation to restore the injectivities of these wells after drilling and completion (Meshal, et al., 2010). In addition, performing this type of work on the wells having ID restrictions posed another challenge in selecting the proper coiled tubing size for open hole acid stimulation application.

Extended reach wells (ERWs) refer to the wells drilled with a trajectory designed to reach distant targets, typically several miles away horizontally from the surface location. One of the typical reasons for these types of wells, ERWs, is associated to surface restrictions or surface facility installation placement which are located within a significant distance of the target entry of the wells. From the standard oil field criteria, any well with the ratio of measured depth to the true vertical depth (MD/TVD) that is greater than 2.0 will

be considered as an extended reach well. From the definition, the ERW can have long measured depth and relatively short and shallow vertical depth (Bhalla, January 1995).

Horizontal wells (Fig. 1) are a type of oil or gas wells where the wellbore is drilled horizontally through a reservoir rock formation. These wells are drilled to intersect and access larger portions of reservoirs that might be otherwise difficult to reach with vertical or shorter horizontal wells. By increasing the length of the horizontal section, operators aim to maximize exposure to productive zones, improve well performance, and enhance ultimate recovery. Although several definitions of horizontal wells exist, for the purposes of this paper, horizontal wells are defined as those exhibiting an inclination greater than 80° throughout the production formations. Horizontal wells are used to enhance production from reservoirs which are not being drained by vertical wells efficiently, where usually these wells have permeability streaks in combination with natural fractures. The horizontal well can link the zones of the reservoir that are productive in a single wellbore. (Carden & Grace, 2007).

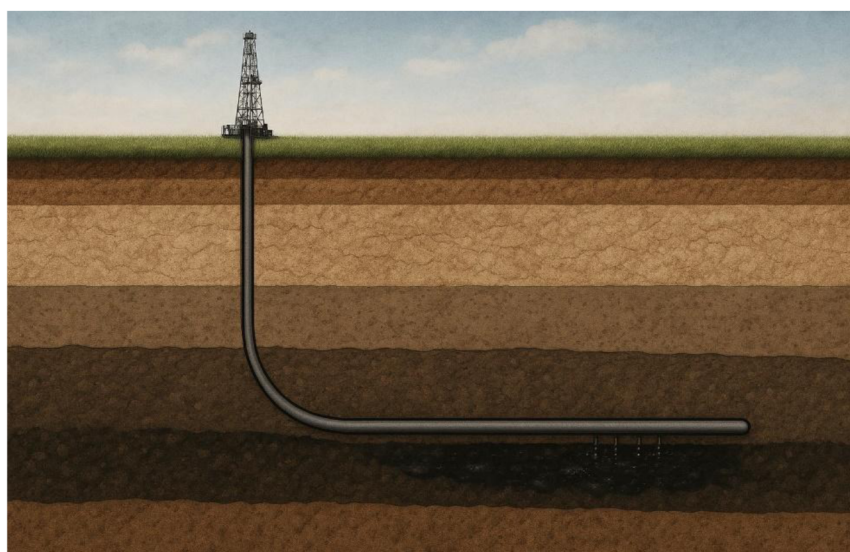


Figure 1—Horizontal well cross-section view

Challenges

One of the most significant challenges in coiled tubing extended-reach applications is the friction and drag along the coiled tubing string as it runs deeper into the wellbore. This friction is dependent on multiple factors like dogleg severity, inclination, and length of the lateral or horizontal section. Since this friction is caused by the contact points of coiled tubing and walls of the hole (open hole or cased hole), so the roughness of the hole is also a major contributing factor. Open hole present higher friction factors compared to cased hole sections. The factors that contribute in this challenge are well depth, horizontal lateral sections, presence of open hole intervals, hole size, and coiled tubing string size. Coiled tubing string size can further be limited due to the following major factors:

- Well architecture (minimum ID restriction of the wellbore/tubing)
- Well trajectory
- Wellbore fluid densities
- Available resources to handle coiled tubing strings on location particularly offshore platform installations.

In acid stimulation and wellbore cleanout operation, extending the reach of coiled tubing to the maximum desired depth or total depth (TD) of the well has significant importance. As the coiled tubing is pushed into the lateral horizontal sections, buckling is induced in the coiled tubing string due to friction and drag. This buckling occurs in stages. The first stage of the induced buckling is called sinusoidal buckling. As the coiled tubing is pushed further from the injector head, this sinusoidal buckling converts to helical buckling. This helical buckling will eventually be converted into lock up where no further coiled tubing movement can be possible (Figure 2). The helical buckling is the key factor to develop lock up of the coiled tubing string as it pushed further into the well bore. Therefore, utilizing the Hydraulic Activated Friction Reduction Tool (HAFRT) to delay helical buckling can effectively prevent lock-up and allow the coiled tubing to reach greater desired depths.

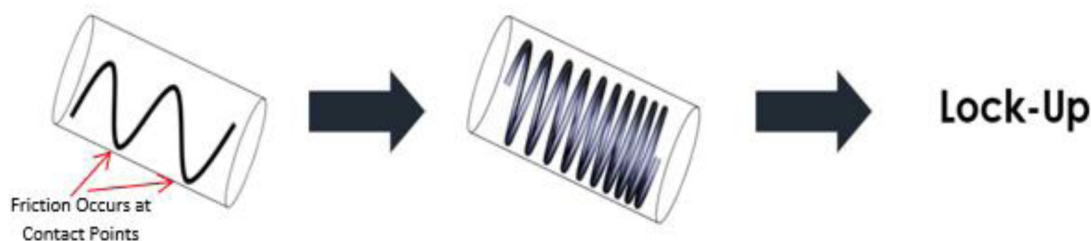


Figure 2—Coiled tubing lock up development

Hydraulic Activated Friction Reduction Tool (HAFRT)

HAFRT is used as a friction reduction tool attached at the end of the coiled tubing string to reduce the drag and frictional forces between the coiled tubing and the wellbore. HAFRT's working principle is to break static friction and convert it to a lower amount of friction i.e dynamic friction. This is achieved by converting fluid hydraulic energy into gentle oscillations that travel axially along the length of the coiled tubing string to reduce the friction between CT and wellbore. This friction is reduced where pressure waves are travelling along the axis of coiled tubing string. This mechanism will result in better improved weight transfer, faster coiled tubing running speed and extending the reach to TD while delaying the buckling of coiled tubing.

HAFRT consists of two major components, power section assembly and valves assembly (Fig. 3). The tool is activated by pumping fluid through the power section which as a result will change the positioning of the valve assemblies. One of the valves is attached to the rotor of power section and the other valve is attached to the bottom side of the tool. Once the fluid is pumped through the tool, the rotor will rotate which will change the orientation and positioning of one valve against the other. By changing this total flow area (TFA), the pressure waves are created which travel axially along the length of the coiled tubing string.

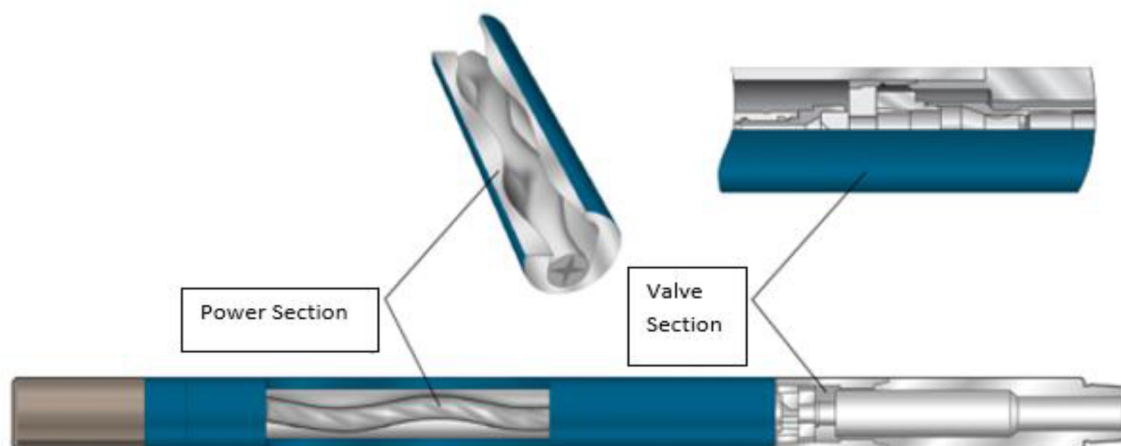


Figure 3—Flow Activated Friction Reduction Tool

Application Study / Implementation / Field Deployment

To perform an acid stimulation operation on a water injector "Well X", several factors are considered to meet the job objectives. In terms of preparation, simulations are crucial for predicting the coiled tubing reach in extended reach horizontal wells to model various factors that are affecting the performance output.

Parameters like wellbore geometry (well survey, well profile), friction factors, coiled tubing string size and operational factors such as flow rates and pressures are key to predicting the coiled tubing reach.

Well Data/ History / Reach Study of "Well X"

Table 1 details the data for "Well X".

Table 1—Well X data

TD (ft)	17,717 ft
Casing Shoe Depth (ft)	11,090 ft
Open Hole Section (ft)	6,627 ft
Open Hole Size (in)	6-1/8"
Well Type	Single Water Injector

Multiple attempts were carried out to perform acid stimulation on "Well X" in the past.

- In 2021, acid stimulation was performed with a maximum reach of 14,141 ft (46% of open hole coverage)
- In 2024, Coiled tubing reach was limited to the maximum depth of 12,757 ft (25% of open hole coverage)

Multiple simulations were performed to study the effect and choose the right coiled tubing string size to achieve maximum reachability for acid stimulation operation.

Table 2 shows the simulation results using 2" in coiled tubing string and the HAFRT.

Table 2—Reach results using 2" CT string

Depth Coverage without HAFRT Open Hole Coverage %	Depth Coverage with HAFRT Open Hole Coverage %
Lock Up Depth 15,149 ft 61%	TD 17,717 ft 100%

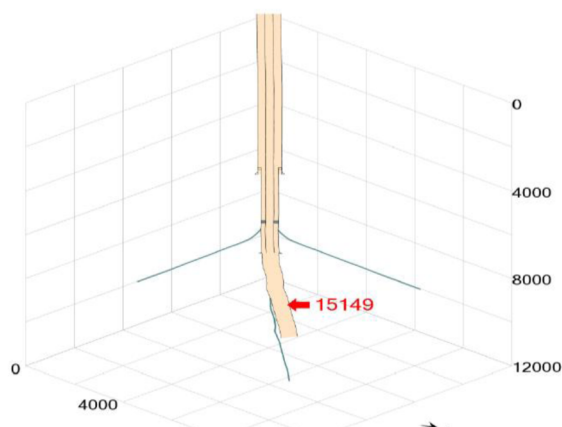


Figure 4—Reach results without HAFR tool

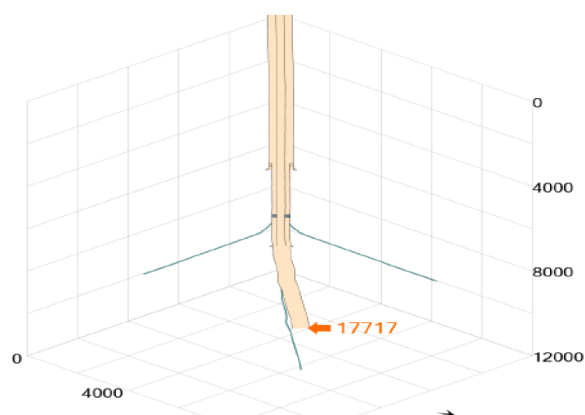


Figure 5—Reach results with HAFR tool

Solution

In order to overcome the challenges associated with minimum ID restrictions (2.67") of the tubing, it was also essential to choose the proper size of the Hydraulically Activated Friction Reduction (HAFR) tool to be used in combination with 2" (in) coiled tubing size.

Hydraulically Activated Friction Reduction (HAFR) was designed to be deployed on 2" coiled tubing string with conventional stimulation bottomhole assembly (BHA). Fig.6 shows the BHA deployed for the planned acid stimulation operation.

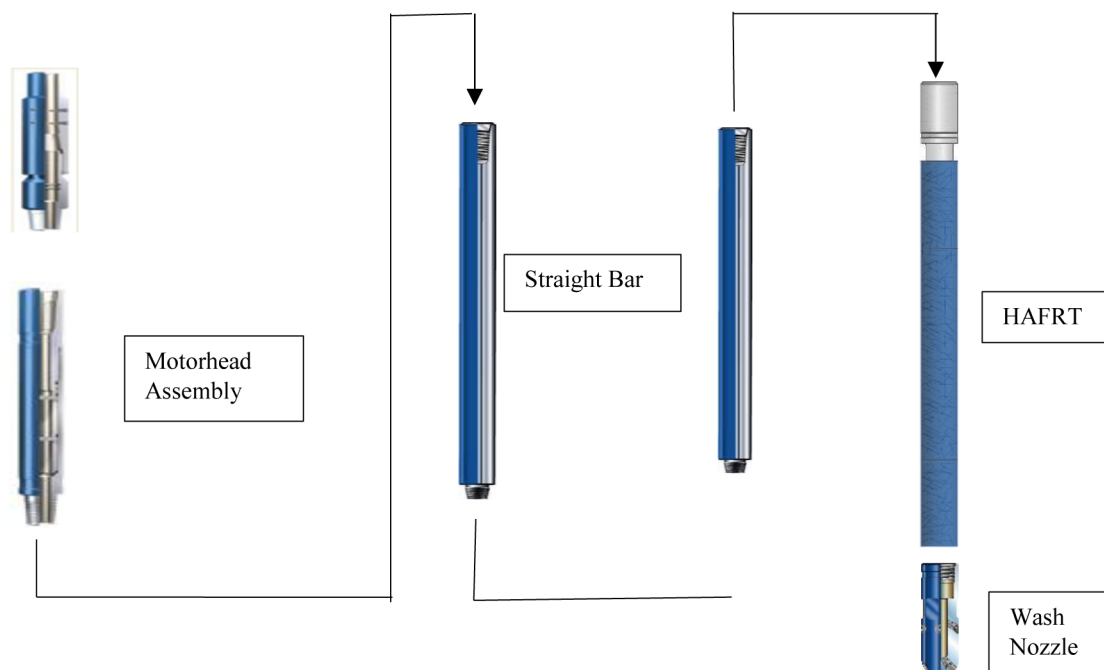


Figure 6—BHA for reservoir cleanup operation

As the fluid flows through, the HAFRT creates gentle oscillation by pressure waves that travel along the axis of coiled tubing string. These oscillations will reduce the friction and contact between the coiled tubing and wellbore. The result of this reduced friction is to improve weight transfer at the end of BHA and to extend the coiled tubing reach to the TD of the well to perform reservoir cleanup operation.

At the beginning of open hole section at 11,090ft, the HAFRT was activated with 1.1 BPM pump rate. As the coiled tubing ran deeper into the well, HAFRT helped to maintain coiled tubing run in hole (RIH) weight to the expected value and overcome any buckling of the CT string due to the drag and friction.

Results

During the run, the following results were achieved using the HAFRT tool.

- Successfully reached the TD of 17,717 ft.
- Prevented early lock up and covered 100% of the open hole interval.
- Entire 100% of open hole interval was stimulated as compared to the previous attempts of maximum coverage of 46%.
- Showed significant consistency in CT RIH weight in actual run as compared to the pre-job simulated CT weight with around 4K lbs more weight at TD.
- Maintained steady progress and CT speed (avg 15-18 ft/min) throughout the open hole section without any lock up.

Comparison of Simulated vs Actual Job Results

Fig. 7 shows the comparison between the simulated RIH CT weight as compared to actual RIH CT weight using HAFRT. There was improvement of 4K lbs more RIH CT weight than simulated results which demonstrates the added value of HAFRT in actual operation.

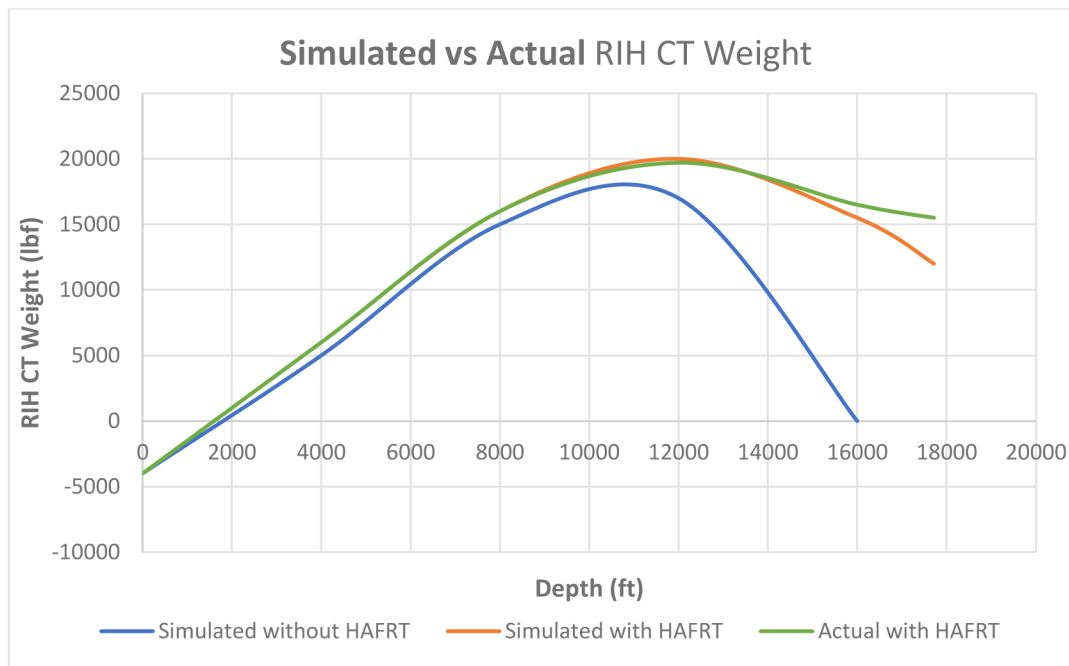


Figure 7—Simulated vs Actual RIH CT Weight

Well Data/ History / Reach Study of "Well Y"

Table 3 details the data for "Well Y".

Table 3—Well Y data

TD (ft)	19,122 ft
Casing Shoe Depth (ft)	10,763 ft
Open Hole Section (ft)	8,359 ft
Open Hole Size (in)	6-1/8"
Well Type	Single Water Injector

No historical accessibility was conducted in this well. However, since the well was commissioned, the injection in the well was drastically reduced and owing to the long horizontal section, the well was considered for stimulation using coiled tubing with deployment of HAFRT tool.

Multiple simulations were performed to study the effect and choose the right coiled tubing string size to achieve maximum reachability for acid stimulation operation.

Table 4 shows the simulation results using 2" in coiled tubing string and the HAFRT.

Table 4—Reach results using 2" CT string

Depth Coverage without HAFRT Open Hole Coverage %	Depth Coverage with HAFRT Open Hole Coverage %
Lock Up Depth 15,264 ft 66%	TD 17,558 ft 100%

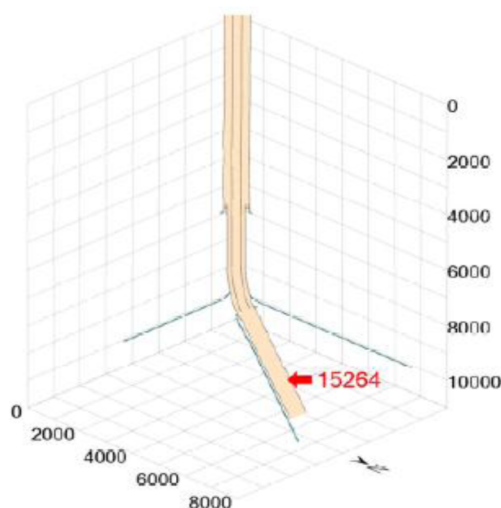


Figure 8—Reach results without HAFR tool

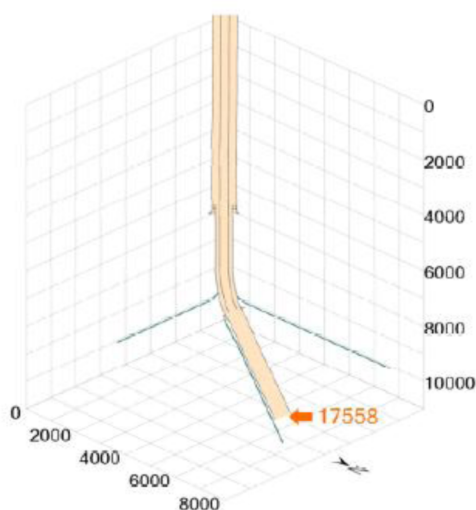


Figure 9—Reach results with HAFR tool

Solution

Well Y being in the same field faces the same challenges of open hole friction and limitation on the coiled tubing size due to ID restrictions. The initial plan was to deploy 2" coiled tubing string but due to availability issues, 1.75" CT string was selected.

The smaller ID of 1.75" CT string limited the HAFRT activation flow rate from the optimum value along with the flexibility to push the coiled tubing as compared to 2" CT string.

Results

During the run, the following results were achieved using the HAFR tool.

- The maximum depth achieved was 15,285 ft.
- Entire 100% of accessed interval was stimulated resulting in 4 folds improvement in injection rate.
- HAFRT tool functioned well however operational limitations related to pumping rate and pumping pressure were encountered due to smaller coiled tubing size.

Conclusion and Recommendations

Hydraulic Activated Friction Reduction Tool (HAFRT) overcame the challenges associated with deploying a bigger sized coiled tubing string due to ID restriction to perform acid stimulation in a challenging open hole environment. The successful deployment of the tool on 2" coiled tubing met the operational objectives while addressing the challenges associated with extended reach and frictional forces. Since 100% accessibility of the open hole was achieved in "Well X", acid stimulation treatment was effectively placed in the entire open hole section as compared to previous multiple attempts of maximum 46% coverage.

On "Well Y", the coiled tubing reach was limited due to smaller sized coiled tubing string (1.75"), which eventually limited the HAFRT activation flow rate. It is recommended to run 2" coiled tubing string for such challenging intervention operation so that HAFRT can be utilized to the full potential in order to meet the operational objectives.

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Nomenclature

HAFRT	: Hydraulic Activated Friction Reduction Tool
CT	: Coiled Tubing
BHA	: Bottom Hole Assembly
TD	: Total Depth
RIH	: Run In Hole
TFA	: Total Flow Area
ERW	: Extended Reach Wells
OH	: Open Hole
BPM	: Barrels Per Minute

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